

The totally-positive Gabor frame-set conjecture is solved

Literature status for arXiv:1104.4894

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Status: literature already answered. Theorem 1 of arXiv:2608.04992 proves the precise frame-set conjecture from arXiv:1104.4894 for every continuous integrable totally positive function and for all real lattice parameters.

Source conjecture

K. Gröchenig and J. Stöckler, *Gabor Frames and Totally Positive Functions*, arXiv:1104.4894, first prove the finite-type case. In the introduction (source lines 204–206) and again in “Remarks and Conjectures” (lines 1133–1140), they formulate the residual conjecture: if $g \in L^1(\mathbb{R})$ is continuous and totally positive, then its Gabor system

$$\mathcal{G}(g, \alpha, \beta) = \{e^{2\pi i \beta l t} g(t - \alpha k) : k, l \in \mathbb{Z}\}$$

is a frame for $L^2(\mathbb{R})$ exactly when $\alpha\beta < 1$.

The paper’s two literal “open problem” extraction signals instead discuss sharp nonuniform sampling and immediately announce the authors’ own finite-type answer. The conjecture above is the distinct, precise residual problem worth checking.

Exact later answer

J. de Dios Pont, K. Gröchenig, L. Liehr, I. Shafkulovska, and M. A. Taylor, *Gabor Frames of Totally Positive Functions: A Complete Characterization*, arXiv:2608.04992 (submitted 5 August 2026), explicitly identifies the earlier frame-set conjecture and says that it is proved in full generality. Its Theorem 1 states

$$\mathcal{F}(g) = \{(\alpha, \beta) \in \mathbb{R}_+^2 : \alpha\beta < 1\}$$

for every continuous totally positive $g \in L^1(\mathbb{R})$, equivalently $\mathcal{G}(g, \alpha, \beta)$ is a frame if and only if $\alpha\beta < 1$. This is an exact hypothesis-and-conclusion match.

Proof architecture checked in the resolving paper

The nontrivial sufficiency direction is reduced by dilation to $\mathcal{G}(g, \alpha, 1)$ with $0 < \alpha < 1$. The paper then:

1. extracts from every coset $x + \alpha\mathbb{Z}$ a sequence $\{k + \delta_k(x)\}$ whose perturbations lie in an interval avoiding the unique zero of the Zak transform and are recurrent under integer shifts;
2. uses total positivity and the de Boor–Friedland–Pinkus criterion to make the associated pre-Gramian $(g(k + \delta_k - l))_{k, l \in \mathbb{Z}}$ surjective on ℓ^1 ;
3. upgrades surjectivity to Fredholmness, while recurrence makes the pre-Gramian a self-similar limit operator;

4. applies the limit-operator injectivity theorem, spectral invariance, and the Gabor submatrix criterion to obtain the frame property.

The standard density obstruction gives necessity. The paper also supplies a Lean formalization of the new sufficiency direction. Thus this packet records literature priority and makes no novelty claim.